

Geoengineering: The Next Climate Disaster?

To many of the younger generation, it feels like none of our policymakers care about our environment. Headlines are filled with increasingly devastating hurricanes, prevalent wildfires, and soaring temperatures, yet many of our legislators remain unconcerned. These natural disasters are not a coincidence: they are effects of global warming. But no real action is being taken to suppress climate change. In fact, in November of 2020, President Donald Trump even backed out of the Paris Agreement, an international treaty created to prevent global warming. And why wouldn't he? Doesn't the prospect of a burning Earth excite you too? Since 2025, rather than progress, we have seen major reversals of emissions regulations and rollbacks of climate programs. To put it bluntly, it seems to me and my peers as if our lawmakers are willing to support anything for profit and political power because they'll die before they have to experience the consequences of climate change. However, to the younger generation, global warming is a growing concern. Just over the last eight years, NASA reports that the Earth's temperature has been the highest ever recorded (Maimann). Scientists predict that once global temperatures reach over 2°C of pre-industrial levels, the planet will reach a "tipping point," when all of the damage done to our planet will not be fixable ("What Is Geoengineering"). *The National Oceanic and Atmospheric Administration*, a U.S. government-run scientific agency predicting changes in climate, reports that we are over halfway to this point, currently standing at 1.46°C above the pre-industrial average. While this may not sound like a lot, the last 40 years alone have seen a global surface temperature increase of 1.15°C (Lindsey and Dahlman). This difference too, may sound tiny—but this change in temperature has melted over two million square kilometers of arctic ice (MIT Climate Portal Writing Team). If, as these models predict,

the Earth's temperature rose 1°C in only a decade, the ensuing disaster will be worthy of Noah's Ark. Honestly, this is just a numbers game, and I am a numbers kind of person. Our global temperatures are increasing at an accelerating rate. Anyone could tell you what will happen next.

Instead of pushing for more renewable energy and stronger emissions regulations, certain governments and private investors are turning to geoengineering. From a science standpoint, I was really excited about what these promising geoengineering techniques could offer. Especially because the Trump administration is actively obstructing emission-reduction progress, turning the issue of climate change from a governmental to a capitalist problem didn't seem like a bad idea. Capitalism has its merits in its efficiency and effectiveness at solving pressing issues. The COVID-19 vaccine, for example, while funded by the government, was developed and tested by private corporations. If geoengineering solutions were similarly promoted by the government and implemented by private companies, it seemed to have the potential to revolutionize climate mitigation (the reducing of climate change). However, the longer I've spent looking into geoengineering, the more I've realized how disproportional the dangers are compared to its potential. These emerging technologies pose not only practical limitations and concerns, but also ethical ones. To me, the more I look into geoengineering, the more it begins to appear as a mere distraction from real climate mitigation.

So what is geoengineering exactly? Geoengineering is the "large or planetary-scale interventions in the Earth's atmosphere, oceans, and soils with the intention of counteracting only some of the effects of climate change" ("Geoengineering"). Although risky, some scientists believe geoengineering implementation is necessary to prevent the planet from reaching this point of no return. As such, geoengineering solutions could serve as a "band-aid," buying us time while we switch to large-scale clean energy (Heutel). Yet others regard geoengineering as a

cheaper, faster, and more effective solution to climate change than traditional mitigation, and promote it enthusiastically. Geoengineering falls mainly into two categories: Carbon dioxide removal and solar radiation modification. While these techniques may vary, after doing a deep dive into the pitfalls of each, I've come to believe that the majority of these shortcuts fail to consider the large-scale consequences of tinkering with our planet's complex environmental systems.

Carbon dioxide removal can include anything from reforestation to emerging technology such as Bio-Energy with Carbon Capture and Storage (BECCS). The latter aims to grow biomass to absorb atmospheric CO₂, burn it to generate power, then capture and store the resulting carbon emissions permanently underground ("What Is Geoengineering"). At first, this sounded like the perfect solution to me. It seems to offer a way to both remove CO₂ from the atmosphere and generate electricity. But as my wise mother says, if something sounds too good to be true, it probably is. The biggest issue surrounding BECCS is the resources it requires. Land that could be used for agriculture or protected as natural ecosystems would need to be dedicated to growing these monoculture plantations. And that doesn't even touch on the water and fertilizer needed to grow these plants (Currie). A 2018 study concluded that the effectiveness of BECCS projects was "strongly case-specific," ranging from a ratio of 0.5 to 5.7 energy return on investment. With a ratio of 1 being energy neutral, this data shows that implementing BECCS could either consume more energy than it produced or be as efficient as solar panels. While the numbers aren't concrete yet, energy leakages from biomass conversion into fuel sources and road transportation should be expected (Brack). I, too, wanted to believe in the potential of BECCS technology—yet all of the facts prove that it won't be more effective than solar panels or wind

turbines. By focusing on developing unpromising technologies, we self-sabotage the opportunity for proven solutions.

The creation of monocultures, increasing competition for land, diminishing water availability, and reducing ecological diversity are not effects exclusive to BECCS, however. They extend to a very widely accepted form of geoengineering: tree planing. One simple demonstration of these consequences is the growing prevalence of “green deserts” in Brazil. Here, genetically modified eucalyptus trees have been engineered for accelerated growth and are marketed by pulp manufacturer Suzano as a cheap way to remove CO₂ from the atmosphere. However, the creation of these new monocultural plantations in Bahia, Brazil has created “green deserts” where these invasive eucalyptus trees create water scarcity and outcompete native plants. Additionally, the introduction of this new market to the area has increased deforestation of native forests and contaminated local communities with agROTOXINS (McClure et al.). While planting trees may seem implicitly beneficial, in reality, they give corporations control over our ecosystems by framing money grabs as carbon assets. Furthermore, the caatinga dry tropical forests found in Bahia, Brazil are already highly-efficient carbon sinks—something that consumes more carbon than it produces—to the extent that they are more efficient than even the Amazon rainforest (Mendes et al.). Suzano is therefore choosing to endanger environmentally crucial carbon sinks for the sake of profitable and ecologically disruptive monocultures. Situations like these prove that the only way to mitigate climate change is to target the root cause, and not by allowing corporations to create their own “fixes.”

However, small companies have been experimenting with another form of geoengineering: solar radiation modification (SRM). SRM is the more publicized, controversial form of geoengineering. It’s often theoretical, although some small-scale experiments are taking

place. The goal of SRM is to cool the Earth by decreasing the amount of sunlight absorbed. This is achieved through increasing the planet's albedo, or reflectiveness. The scientific consensus is that increasing the Earth's albedo will decrease surface temperatures, as proven through ice-albedo feedback: as ice melts, the planet becomes less reflective and warms faster in a snowballing effect (Von Paris et al.). This can be done through increasing the amount of white and reflective surfaces on the Earth, whether that be through increasing cloud cover, ice cover, or even painting urban rooftops white. The startup "Stardust Solutions" has already begun experimentation with stratospheric aerosol injection (SAI), a geoengineering method inspired by the cooling effect of the reflective sulfur dioxide particles ejected in major volcanic eruptions. Using balloons, the company releases these particles into space, intending to mimic this natural cooling process (Kolbert). Theoretically, implementing technology directly imitating Earth's natural processes seems relatively safe.

But while it may seem appealing to complete small-scale SRM now so it becomes a plausible option in the future, geoengineering research poses complex and significant risks. Geoengineering research cannot be contained in a lab, and instead modifies our complex and interconnected climate system directly. Because of the climate's numerous feedback loops, the majority of effects will silently ripple through our global ecosystem, only becoming apparent once it's too late. One proposed SRM approach comes from scientists at Dartmouth College and Yale University. They suggest that while boreal trees in Canada extract CO₂ from the air, clearing these forests could actually have a net cooling effect, as the area's snow coverage would increase and improve the Earth's albedo (Chalmin). This is because snow reflects light back, rather than absorbing it like the dark green boreal foliage. This kind of tradeoff complicates the logic of geoengineering, and when I first encountered this idea, I was in disbelief. It felt like common

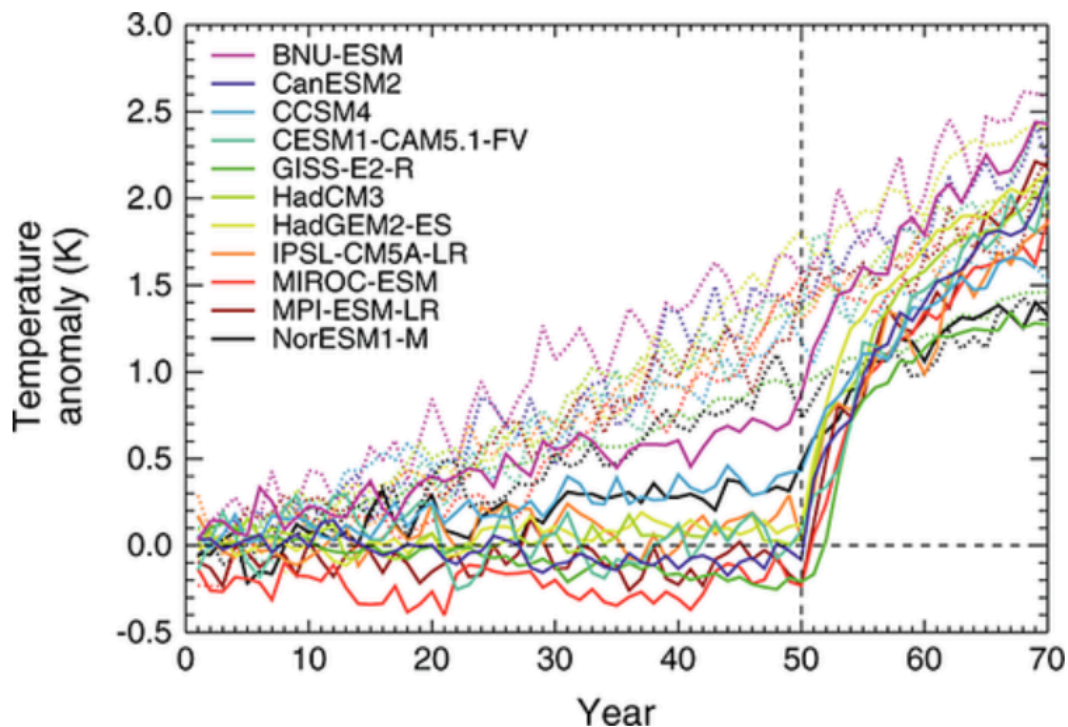
sense to reject a mitigation strategy that relies on destroying huge organisms which absorb carbon emissions in favor of geoengineering strategies which don't reduce carbon at all.

This skepticism led me to examine additional resources on the issue. It was no surprise to me, therefore, when I found other climate models that suggested quite the opposite. A 2025 study published in *Nature*, an international peer-reviewed journal on science and technology, advised that planting between 6.4 to 32 million hectares of trees on Canada's northern boreal forest edge could extract between 3.88 and 19.4 gigatonnes of carbon dioxide over 75 years—that's over five times Canada's annual greenhouse gas emissions (Dsouza). But I also think it would be overly simplistic to only consider surface albedo when it comes to net cooling, as our ecosystems are more complicated than that. A 2024 study supports this theory, suggesting that planting trees in high latitude areas would not only decrease planet albedo, but also disturb the solid carbon stored in soil, releasing the CO₂ back into the air (Kristensen et al.). Due to the complex relationships between these forest ecosystems, causing a change in forest cover can have unpredictable impacts on cloud coverage, evapotranspiration, and snowmelt in spring (Chalmin). These latter two studies illustrate how the climate is literally a labyrinth of interconnected systems. As it stands, our data is contradictory and extremely unreliable.

This nature of unpredictability is not isolated to reforestation on Canada's northern boreal forest edge. Countless computer models have simulated the effects of solar geoengineering, each producing more and more unexpected results. In a 2022 study published by the *Nature* journal, SAI was predicted to alter global precipitation patterns and reduce agricultural productivity (Asutosh et al.). Another study published by the *European Geosciences Union* suggests that SAI could cause a decrease in precipitation over southern Europe and create "permanent westerly winds in the lower stratosphere" (Jones et al., "Stratospheric Aerosol"). Each study predicts

slightly different effects—yet they all agree that this technology will cause a fundamental shift in global weather patterns. A 2023 study even says that the climate impacts of SAI are more dependent on the computer model used than anything actually modeled, making its outcomes difficult to predict (Fasullo and Richter). All in all, that is to say we have no idea what will happen after SAI implementation. The last thing we should begin is small-scale testing.

The good thing is that SAI is not currently being implemented on a large scale. However, climate model simulations show that if SAI were implemented and then abruptly stopped after decades of deployment, global temperatures would soar in a phenomenon known as termination shock. This is the result of decades of suppressed warming taking effect over the course of a few years. This effect is demonstrated in a model created by Dr. Anthony Jones, a scientist working in the Met Office to forecast climate. His 2017 paper shows the predictions of various computer models for before and after solar radiation modification is stopped (Jones et al., “Abrupt Suspension”).



Each colored and dashed line represents a different computer model's prediction of Earth's rising temperatures. Temperatures during solar aerosol injection implementation are indicated to the left of the vertical dashed black line, while temperatures after the injections are abruptly stopped are indicated to the right of the line. The horizontal dashed line represents what is considered a healthy temperature (Jones et al., "Abrupt Suspension"). All computer models unanimously support that if solar aerosol injections were abruptly stopped, Earth's temperatures would soar no less than one degree kelvin (or one degree Celsius) in a decade. This termination shock shows how dangerously irreversible SAI is, meaning that any large-scale implementation would commit future generations to maintaining it.

Even if—against all odds—governments manage to sustain that level of coordination, it is also imperative to note that SAI, among other SRM techniques, is not a catch-all solution to climate change. The problem is that the novelty of geoengineering captivates both scientists and climate advocates alike as a solution to climate change, despite not being a solution at all. The underlying problem to global warming is increasing CO₂ emissions, caused by fossil fuel vehicles, industrial plants, and famously, cow farts. In the 50-year old *Science for the People* magazine, Linda Schneider, a climate policy expert at Heinrich Boll Foundation, writes extensively about geoengineering from a climate justice perspective in her article, "Geoengineering and Climate Capitalism." She illustrates how SRM would have no effect on any CO₂ emitters. SRM treats the symptoms of climate change while ignoring the disease (Schneider). The increase of CO₂ in the atmosphere has been partially absorbed by the ocean, causing ocean acidification—an effect upsetting the pH balance of the Earth's ocean ecosystems—which also remains unaffected by proposed SRM solutions (Heutel). SRM ignores the root cause of climate change and only treats the byproducts directly related to temperature

increases—reducing cumulative CO₂ emissions through other means is necessary for effective climate mitigation.

While geoengineering poses practical problems, a larger concern may just be its moral implications when it comes to issues of fault and responsibility. By supporting empty geoengineering solutions which are proven ineffective, corporations and industries are able to adopt a front of sustainability while avoiding accountability for the climate crisis. Schneider also suggests that “were the international community to address the root causes of environmental destruction, major pollutants would bear the political and economic costs” (Schneider). Rather than focusing on decreasing emissions, large corporations are able to promote geoengineering while continuing business as usual. Another example of this is fossil fuel companies’ promotion of biochar. Biochar is a CO₂ removal method where carbon waste from agriculture is heated into charcoal and buried to store carbon in the soil. While again, this geoengineering technology appears promising on the surface, an article published in the *NASA Astrophysics Data System* illustrates the logistical challenges of large-scale biochar implementation, which keeps its application limited to within a laboratory (Wani). Despite biochar’s current unscalability, Steve Horn, a climate investigator reporter, reveals that powerful fossil fuel companies have previously lobbied to include biochar in climate policy (Horn). Essentially, through their support, emitters of greenhouse gases are able to buy cooling credits from projects that remove carbon, thus claiming to become more sustainable without actually reducing their own pollution (Chalmin). When carbon emissions become tradable units, the illusion of progress becomes all too easy to fake. Through carbon markets, corporations “buy their right” to pollute. This protection of the oil and gas industry’s emissions is consistent with their broader stance on climate policy. From 2000 to 2016, oil companies have spent a total of \$370 million supporting climate change denial and

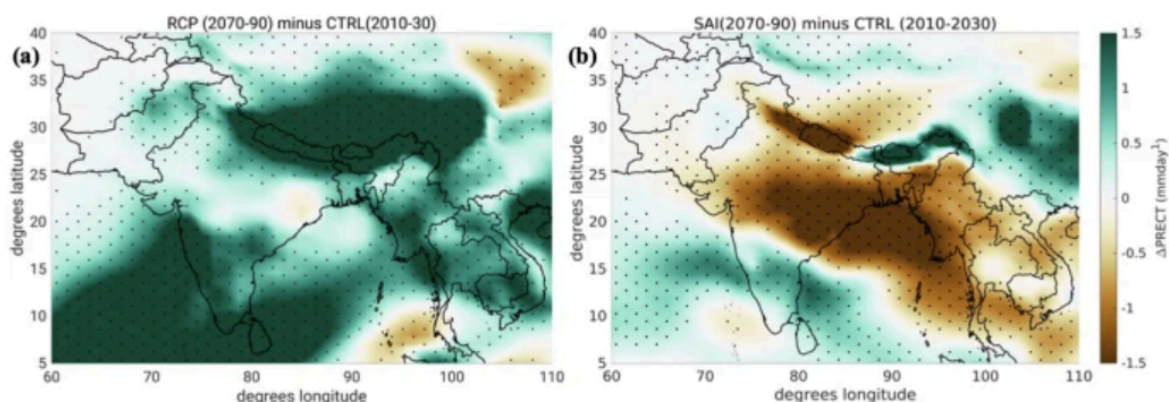
lobbying the U.S. government on climate change (Heninger). The oil and gas industry has proven time and time again that it will support anything to avoid real action today and shift that responsibility onto future promises. Promoting geoengineering is simply part of that plan.

What really gets me is that instead of owning up to their actions or even just destroying the Earth outright, these companies deliberately trick their consumers through supporting “eco-friendly” initiatives. And well-meaning people fall for it because they *want* to support ethical companies. For example, my dad will exclusively buy free-range chicken eggs to ensure he’s supporting a more humane treatment of animals. But by definition, a fossil fuel company *cannot be* “eco-friendly”: their business is extractive and reliant on the continued production and combustion of the very thing driving climate change. Through “buying their right” to pollute, the gas and oil industry tricks conscious consumers into not only believing in unpromising technologies, but also in their ethicality as an industry. Nonetheless, for many, these false initiatives are what introduces them to the idea of climate activism. While action is better than inaction, at the end of the day, all this does is expand misinformation and divert attention away from proven climate mitigation solutions.

The ethical ramifications of pursuing geoengineering doesn’t end with shifting blame, however. The ability to perform geoengineering research is in itself an issue. The broad environmental risks of geoengineering create power imbalances between the select countries doing geoengineering and all of the countries affected. Now we have to contend with the issues of environmental justice. If actions in one country may inadvertently affect the weather in another, how can multiple nations collaborate? Who is allowed to decide what projects to implement? While wealthy individuals or small countries may be able to conduct small-scale experiments, full-scale SRM implementation requires the resources of large nations (Schneider).

In a 2025 study published in *Nature*, A. Asutosh, a researcher at the National Center for Atmospheric Research, created computer models that show how SAI may reduce rainfall during the summer monsoon season in northern and central India. Although fewer monsoons may seem like a good thing, this season is incredibly important for the agriculture and ecosystems that support nearly two billion people (Asutosh et al.).

Fig. 1: SAM precipitation under RCP and SAI.



The figure on the left is of precipitation in a simulation where the representative concentration pathway (RCP) is 8.5, indicating high emissions, and no geoengineering. In this future, rainfall is expected to increase near the Bay of Bengal. The rightmost image shows the same area under SAI, demonstrating less rainfall over the Bay of Bengal (Asutosh et al.). While it's true that these computer models may not be 100% accurate, when considering the effects of one of the greatest global crises of the century, any predicted risks should be taken seriously. The problem is that the majority of geoengineering experiments and research is happening in the Global North, while some predicted effects occur in the Global South. Solar geoengineering implementation such as SAI would violate the state sovereignty of other nations by forcing them to deal with unconsented side effects in their nations. Even if countries such as India decide to not involve themselves with SAI, should any other country choose to deploy it, the effects of SAI

would be unavoidable. Indigenous people face a similar issue around consent. Indigenous people are reliant on vulnerable native ecosystems, and are therefore more affected by the consequences of geoengineering (Currie). Tom Goldtooth, of the Indigenous Environmental Network, said that to protect the “distinct legal, political, inherent, and collective rights” of indigenous people, geoengineering should not be used as a solution to climate change ("Press Release"). If large scale geoengineering is implemented, it is clear that someone will pay the price. Whether they are invisibly suffering on the sidelines or directly in the line of fire, the consequences of geoengineering are unavoidable for those without a voice in this research field.

Similarly, power imbalances feed into climate colonialism, where companies in the global north choose to export the risks of local geoengineering experiments onto places like Africa to “cut costs.” Dr. Mfoniso Xael, of the Health of Mother Earth Foundation said,

The idea of turning Africa into a laboratory for dangerous, unproven technologies like solar radiation manipulation is outrageous and insulting...Carbon markets and offsets are nothing but greenwashed camouflage that allows polluters to keep polluting while exploiting our land and people...Africa is not your experiment, not your carbon sink, and not for sale. (qtd. in "Press Release")

While places like Africa did not create the climate crisis, we allow them to eat the risks involved with “fixing” it. All geoengineering solutions require infrastructure and resources such as water, energy, and land, making competition over these resources unavoidable (Schneider). When the U.S. uses its global influence and industrial might to exploit the resources of underdeveloped nations—ones which have contributed least to climate change and often have less decision-making power—we, too, are avoiding responsibility.

These same inequalities create another danger: geoengineering, although designed as a mitigation tool, has the potential to be weaponized by powerful actors. These fears are not unwarranted. The existence of research often justifies deployment in warfare. Take nuclear weapons as a precedent. During World War II, the development of nuclear weapons created a new strategic option. They provided a path to save American lives and shorten the war, thus justifying their usage, disregarding the millions of Japanese civilian lives ended by this decision. The use of nuclear weapons during WWII proves that the path from research to deployment is a very slippery slope. Once the technology becomes available, deployment becomes the quicker and easier option. Just as scientists in 1945 didn't understand the extensive effects of nuclear fallout, today we still don't quite know what effects SAI or deforestation may have on our planet. Rather than pursuing sustainable climate mitigation, we would always be tempted by this “easy way out.”

Geoengineering not only fails to address the root causes of climate change, but also poses significant risks to international communities through unintended effects on the global climate, threats to certain ecosystems, and the risk of weaponization—often without the knowledge or consent of the communities affected. Geoengineering has undeniable practical and ethical concerns. However, it would be unrealistic to suggest that the advancement of any technology can just be halted—especially not geoengineering, where the shiny novelty of these high-tech solutions are simply too alluring in their promises to “solve” the climate crisis. As part of the human race, we chase progress—even if it could lead us to our doom. Therefore, should geoengineering experimentation continue regardless, legal and ethical frameworks are essential to safeguard its continued development. As it currently stands, there is a legal void around who is held liable should any experiments go wrong. Such governance is needed to manage the risks

and ensure equality in a geoengineering future. This means creating global treaties, outlining international laws, and agreeing on ethical codes to keep each other accountable. There is no perfect solution to the climate crisis. Geoengineering is certainly far from being one.

The thing is, we are not helpless. By holding fossil fuel companies accountable for their performative sustainability efforts and the smog they continue to pump into our atmosphere, we can move away from fossil fuels. You can limit your own carbon footprint by driving less or installing solar panels on your roof. You can vote for candidates who want to limit fossil fuel expansion. Even just educating yourself on climate mitigation and sharing that knowledge with others is effective. Additionally, renewable energies such as solar and wind power are competitive with gas and coal. If we direct our efforts toward developing renewable technologies instead of funding SAI and use land space for sustainable energy implementation rather than monoculture forests, we can commit to real climate mitigation. We can also support naturally existing carbon sinks by campaigning against illegal logging, building infrastructure over wetlands, protecting grasslands, and saving our oceans from acidification. These efforts will undeniably take time and significant climate advocacy. I am not denying how these are ambitious goals. However, switching to sustainable energy sources and supporting natural carbon sinks is the safest and most ethical path forward.

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